

ELEC 7410 Homework Assignment #7 Nov. 6, 2025

(6 problems, Due: Nov. 13, 2025)

Any late homework submission will incur 15% penalty per day, with any fraction of a day counted as full day.

1. (10 points) (**Characterization of random processes**) Problem 5, Chapter 10, Gubner.

5. Let X_t for $t > 0$ be a random process with zero mean and correlation function $R_X(t_1, t_2) = \min(t_1, t_2)$. If X_t is Gaussian for each t , write down the probability density function of X_t .

2. (10 points) (**Strict-sense and Wide-sense Stationary Processes**) Problem 16, Chapter 10, Gubner.

16. A discrete-time random process is WSS if $E[X_n]$ does not depend on n and if the correlation $E[X_n X_m]$ depends on n and m only through their difference. In this case, $E[X_n X_m] = R_X(n - m)$, where R_X is the univariate correlation function. Show that if X_n is WSS, then so is $Y_n := X_n - X_{n-1}$.

3. (10 points) (**Characterization of correlation functions**) Problem 52, Chapter 10, Gubner.

*52. Let $R(\tau)$ be a correlation function, and for fixed $\tau_0 > 0$ put

$$\bar{R}(\tau) := R(\tau - \tau_0) + R(\tau + \tau_0).$$

Select the best answer from the following (justify your choice):

- (a) $\bar{R}(\tau)$ is always a correlation function.
- (b) For some choice of $R(\tau)$ and $\tau_0 > 0$, $\bar{R}(\tau)$ is a correlation function.
- (c) There is no choice of $R(\tau)$ and $\tau_0 > 0$ for which $\bar{R}(\tau)$ is a correlation function.

4. (10 points) (**The Wiener Process**) Problem 25, Chapter 11, Gubner.

25. Let V_t be defined as in Example 11.6. Show that for $0 \leq s < t < \infty$, $E[V_t V_s] = \sigma^2 s$.

5. (10 points) (**The Wiener Process**) Problem 27, Chapter 11, Gubner.

27. Let W_t be a Wiener process with $E[W_t^2] = \sigma^2 t$. Put $Y_t := e^{W_t}$. Find the correlation function $R_Y(t_1, t_2) := E[Y_{t_1} Y_{t_2}]$ for $t_2 > t_1$.

6. (10 points) (**The Wiener Process**) Problem 28, Chapter 11, Gubner.

28. Let the random vector $X = [W_{t_1}, \dots, W_{t_n}]'$, $0 < t_1 < \dots < t_n < \infty$, consist of samples of a Wiener process. Find the covariance matrix of X , and write it out in detail as

$$\begin{bmatrix} c_{11} & c_{12} & c_{13} & & c_{1n} \\ c_{21} & c_{22} & c_{23} & & c_{2n} \\ c_{31} & c_{32} & c_{33} & & c_{3n} \\ & & & \ddots & \\ c_{n1} & c_{n2} & c_{n3} & & c_{nn} \end{bmatrix},$$

where each c_{ij} is given explicitly in terms of t_i or t_j .